

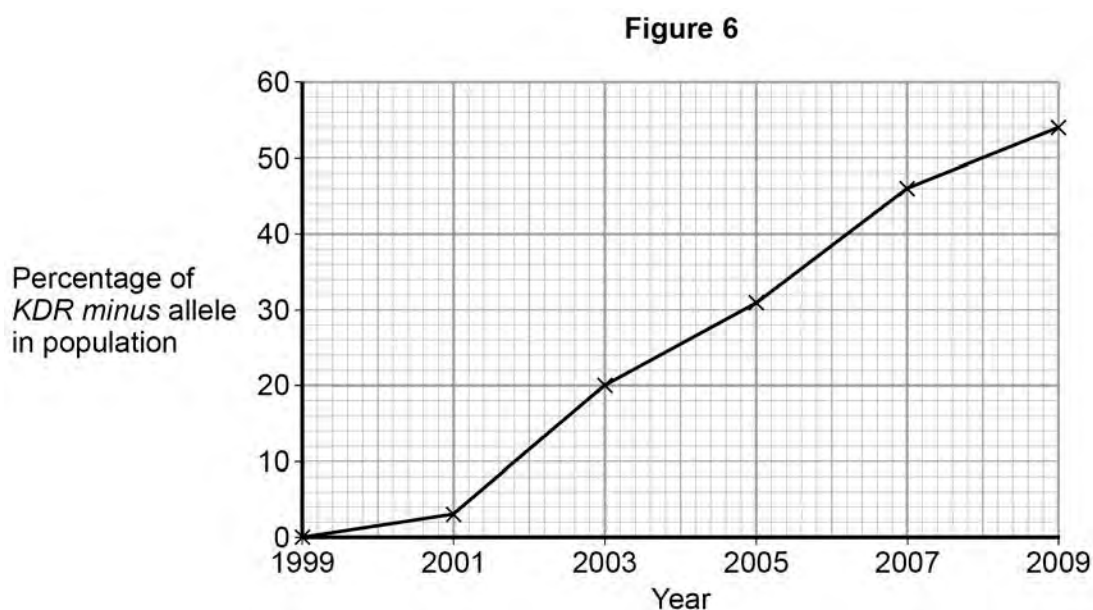
6

Malaria is a disease that is spread by insects called mosquitoes. In Africa, DDT is a pesticide used to kill mosquitoes, to try to control the spread of malaria.

Mosquitoes have a gene called *KDR*. Today, some mosquitoes have an allele of this gene, *KDR minus*, that gives them resistance to DDT. The other allele, *KDR plus*, does not give resistance.

Scientists investigated the frequency of the *KDR minus* allele in a population of mosquitoes in an African country over a period of 10 years.

Figure 6 shows the scientists' results.



06

. 1

Use the Hardy–Weinberg equation to calculate the frequency of mosquitoes heterozygous for the *KDR* gene in this population in 2003.

Show your working.

[2 marks]

Frequency of heterozygotes in population in 2003 _____

0 6 . **2** Suggest an explanation for the results in **Figure 6**.

[4 marks]

[Extra space]

The *KDR plus* allele codes for the sodium ion channels found in neurones.

0 6 . **3** When DDT binds to a sodium ion channel, the channel remains open all the time. Use this information to suggest how DDT kills insects.

[2 marks]

0 6 . **4** Suggest how the *KDR minus* allele gives resistance to DDT.

[2 marks]

